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\documentclass[10pt,twocolumn]{article}
\usepackage[letterpaper,top=0.62in,bottom=0.55in,left=0.6in,right=0.6in]{geometry}
\usepackage{times}
\usepackage{amsmath,amssymb}
\usepackage{graphicx}
\usepackage{booktabs}
\usepackage{caption}
\usepackage{algorithm}
\usepackage[noend]{algpseudocode}
\usepackage[hidelinks]{hyperref}
\usepackage{balance}

\captionsetup{font=small,labelfont=bf,skip=6pt,justification=justified}
\captionsetup[figure]{position=below}
\setlength{\textfloatsep}{11pt plus 2pt minus 2pt}
\setlength{\floatsep}{11pt plus 2pt minus 2pt}
\setlength{\intextsep}{11pt plus 2pt minus 2pt}
\setlength{\abovecaptionskip}{6pt}
\setlength{\belowcaptionskip}{2pt}
\setlength{\columnsep}{0.26in}
\renewcommand{\baselinestretch}{1.0}
\pagestyle{empty}

\renewcommand{\thesection}{\Roman{section}}
\renewcommand{\thesubsection}{\Alph{subsection}}
\makeatletter
\renewcommand{\section}{\@startsection{section}{1}{\z@}{8pt plus 2pt minus 1pt}%
  {3pt}{\centering\normalsize\bfseries}}
\renewcommand{\subsection}{\@startsection{subsection}{2}{\z@}{6pt plus 1pt minus
  {2pt}{\normalsize\itshape}}
\makeatother

\algnewcommand\algorithmicrequire{\textbf{Input:}}
\newcommand{\MLKEM}{ML-KEM-768}

\begin{document}

\twocolumn[
\begin{center}
{\large\bfseries ML-KEM-Based Quantum-Resilient Checkpoint File Transfer\}[2pt]
Protocol for 6G Networks\par}
\vspace{7pt}
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\vspace{3pt}

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 $\end{center}$

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$\begin{minipage}{\textwidth}$

$\small$

$\noindent\textit{\textbf{Abstract}}$ ---Millimetre-wave 5G access already suffers f

$\vspace{4pt}$

$\noindent\textit{\textbf{Keywords}}$ ---post-quantum cryptography, ML-KEM, FIPS 203

$\end{minipage}$

$\vspace{13pt}$

]

$\section{INTRODUCTION}$

Link disruption is not a future problem. 5G mmWave access already suffers frequen

Conventional key agreement (RSA, ECDH) falls to Shor's algorithm, and traffic rec

$\subsection{Related Work}$

Recent work on post-quantum wireless security falls into three groups. The first

$\subsection{Contributions}$

$\begin{enumerate}\itemsep1pt \parskip0pt \topsep2pt \leftskip-5pt$

$\item$ A protocol combining MLKEM key establishment, AES-256-GCM bulk protectio

$\item$ A safety argument for resume: a deterministic, chunk-index-derived GCM nonc

$\item$ A closed-form delay model and the checkpoint interval $\$N^{\ast}$ minimising pe

$\item$ A six-metric performance evaluation at one operating point for each of 5G,

$\end{enumerate}$

$\section{SYSTEM MODEL}$

$\label{sec:model}$

$\subsection{Network and Disruption Model}$

Sender and receiver are endpoints (UEs, or a UE and an edge server) whose path tr

$\subsection{Cryptographic Model}$

A file of $\$S\$$ bits is split into $\$n_c=\lceil S/c\rceil\$$ chunks of $\$c\$$ bits. The s

$\begin{equation}$

$T_h = t_{\text{kg}} + t_{\text{enc}} + t_{\text{dec}} + \frac{|pk| + |ct|}{R} + \tau,$

$\end{equation}$

which is RTT-dominated, since the lattice operations are microsecond-scale. Keys

$$K_{\text{ck}} = \text{HKDF-SHA256}(\text{ss}, \text{sid}, \text{pq6g/v1''}),$$

where K is the AEAD key and K_{ck} keys the checkpoint MAC~\cite{hkdf}. Chunk

$$C_i = \text{AES-256-GCM}_{K_{\text{ck}}}(\nu_i, M_i, A_i), \quad \nu_i = \text{sid}_{32} \parallel i,$$

with associated data $A_i = (\text{sid}, i, |M_i|)$ ~\cite{gcm}. Because ν_i is a pure fu

$$\chi = \text{Big}(\text{sid}, i^*, o^*, H_{i^*}, \text{HMAC}_{K_{\text{ck}}}(\text{sid} \parallel i^*))$$

written durably by the receiver after verifying each chunk's tag. The sender neve

Layer & Component			
Key establishment & MLKEM{} (FIPS 203), liboqs/kyber-py			
Key schedule & HKDF-SHA256 $\rightarrow$ AEAD + checkpoint MAC key			
Bulk encryption & AES-256-GCM, deterministic nonce $\text{sid} \parallel i$			
Transport & TCP sockets, custom framed protocol			
Channel emulation & token bucket, jitter, Poisson blockage; netem			
Inspection / eval. & Wireshark + Lua dissector; 3-arm harness			
File / chunk / t_{ck} & 100 MB / 64 KB / 5 ms, interval $N=64$			
Parameter & 5G mmWave & Beyond-5G & 6G THz			
Link rate R	& 800 Mbps	& 900 Mbps	& 1 Gbps
Round-trip time τ	& 2 ms	& 1.5 ms	& 1 ms
Detect / reconnect	& 0.3 / 0.3 s	& 0.4 / 0.4 s	& 0.5 / 0.5 s
Blockage rate λ	& 0.8 s^{-1}	& 1.0 s^{-1}	& 1.2 s^{-1}

begin{figure*}[t]

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\centering
\includegraphics[width=0.93\textwidth]{figs/fig1.pdf}
\caption{System architecture. Sender and receiver run symmetric crypto, checkpoin
\label{fig:arch}
\end{figure*}

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\section{PROPOSED FRAMEWORK}

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\subsection{Architecture}

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Fig.~\ref{fig:arch} shows the architecture: the sender splits the input into sequ

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\subsection{Control Flow and Algorithm}

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Fig.~\ref{fig:flow} traces the same logic as a state machine. The upper row is th

```

\begin{algorithm}[t]
\small
\caption{Checkpointed quantum-resilient transfer (sender)}
\label{alg:send}
\begin{algorithmic}[1]
\Require file  $F$ , receiver address, chunk size  $c$ , interval  $N$ 
\State  $(pk) \backslash \text{gets} \backslash \text{textsc{Recv}} (\backslash \text{textsc{Hello}})$ ;  $(ct, ss) \backslash \text{gets} \backslash \text{text{MLKEM.Encaps}}$ 
\State  $\backslash \text{textsc{Send}} (ct)$ ;  $\$K \backslash K_{ck} \backslash \text{gets} \backslash \text{text{HKDF}} (ss, sid)$ 
\State split  $F$  into  $M_0 \ldots M_{n_c-1}$ ;  $\$i \backslash \text{gets} 0$ ;  $\$H_{-1} \backslash \text{gets} \backslash \text{varnothing}$ 
\While{ $\$i < n_c$ }
\State  $\$nu_i \backslash \text{gets} sid \backslash i$ ;  $\$C_i \backslash \text{gets} \backslash \text{textsc{Seal}}_K(\backslash nu_i, M_i, A_i)$ ;  $\backslash \text{textsc{H}}_i \backslash \text{gets} H(H_{i-1} \backslash H(C_i))$ 
\If{ $\backslash \text{textsc{Ack}} (i)$  received}
\State  $\$H_i \backslash \text{gets} H(H_{i-1} \backslash H(C_i))$ 
\If{ $\$i \bmod N = 0$ } \State  $\backslash \text{textsc{Persist}} (\backslash chi)$  \EndIf
\State  $\$i \backslash \text{gets} i+1$ 
\ElseIf{disruption detected}
\State  $\backslash \text{textsc{Pause}}$ ; reconnect with exponential backoff
\State  $\backslash chi \backslash \text{gets} \backslash \text{textsc{RequestResume}} (sid)$ 
\State abort if  $\backslash \text{text{HMAC}}_{K_{ck}}(\backslash chi)$  invalid or  $\$H_{i^*}$  mismatche
\State  $\$i \backslash \text{gets} i^*+1$ 
\EndIf
\EndWhile
\State  $\backslash \text{textsc{Send}} (\backslash \text{textsc{Fin}}, \backslash \text{text{SHA-256}} (F))$ ; verify receiver digest
\end{algorithmic}
\end{algorithm}

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\begin{figure*}[t]
\centering
\includegraphics[width=0.93\textwidth]{figs/fig2.pdf}
\caption{Control flow. The recovery path (lower row) re-enters the transmit loop
\label{fig:flow}

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\end{figure*}

\subsection{Optimal Checkpoint Interval}

With n_d expected disruptions, the total overhead of persistence plus expected

\begin{equation}

$$O(N) = \underbrace{\frac{S}{Nc}}_{\text{persistence}} \cdot \frac{1}{t_{ck}} + \underbrace{n_d}_{\text{disruptions}} \cdot \frac{1}{t_{ck}}$$

\end{equation}

convex in N ; setting $\frac{d}{dN} O(N) = 0$ gives

\begin{equation}

$$N^* = \sqrt{\frac{2RS}{t_{ck}}} \cdot n_d \cdot c^2.$$

\end{equation}

For $S=100$ MB, $c=64$ KB, $R=1$ Gbps, $t_{ck}=5$ ms and $n_d=5$: $N^* \approx 76$

\section{IMPLEMENTATION, RESULTS AND PERFORMANCE ANALYSIS}

\label{sec:results}

The protocol is being implemented in Python 3.10 over TCP sockets with a Wireshar

\subsection{Performance Metrics}

Six metrics are used, each comparable across generations: \textit{(i)} completion

\begin{figure*}[t]

\centering

\includegraphics[width=0.99\textwidth]{figs/fig3.pdf}

\caption{Analytical projection ($S=100$ MB, $N=64$): (a) completion time at the 6

\label{fig:res}

\end{figure*}

\begin{figure*}[t]

\centering

\includegraphics[width=0.99\textwidth]{figs/fig4.pdf}

\caption{Performance metrics across the 5G $\rightarrow$ B5G $\rightarrow$ 6G pro

\label{fig:metrics}

\end{figure*}

\subsection{Results}

At 6G scale, completion time falls by 24.1\% at five disruptions and 26.2\% at te

Applied along the generation progression, the same closed form gives the monotone

Fig.~\ref{fig:res}(c) shows the interval trade-off: flat near the optimum, \textit{t

\begin{table}[t]

\centering

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\caption{Projected performance metrics at five disruptions. B = baseline restart,
\label{tab:metrics}
\footnotesize
\setlength{\tabcolsep}{3.4pt}
\begin{tabular}{lcccccc}
\toprule
& \multicolumn{2}{c}{5G mmWave} & \multicolumn{2}{c}{Beyond-5G} & \multicolumn{2}{c}{} \\
\cmidrule(lr){2-3}\cmidrule(lr){4-5}\cmidrule(lr){6-7}
Metric & B & P & B & P & B & P \\
\midrule
Completion time (s) & 6.63 & 4.24 & 7.23 & 5.11 & 7.92 & 6.01 \\
\quad \textit{reduction} & \multicolumn{2}{c}{36.1\%} & \multicolumn{2}{c}{29.3\%} & \multicolumn{2}{c}{} \\
Goodput (MB/s) & 15.1 & 23.6 & 13.8 & 19.6 & 12.6 & 16.6 \\
Redundant data (MB) & 250 & 10.5 & 250 & 10.5 & 250 & 10.5 \\
Transfer efficiency & 28.6\% & 90.5\% & 28.6\% & 90.5\% & 28.6\% & 90.5\% \\
Recovery / outage (s) & 1.13 & 0.62 & 1.27 & 0.82 & 1.42 & 1.02 \\
Handshake material (B) & 13\,632 & 2272 & 13\,632 & 2272 & 13\,632 & 2272 \\
Checkpoint overhead  $\rho$  & --- & 11.9\% & --- & 13.4\% & --- & 14.9\% \\
\bottomrule
\end{tabular}
\end{table}

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\subsection{Discussion}

\MLKEM{} is computationally cheaper than P-256 ECDH but its $\$pk/ct\$$ are more than

\section{CONCLUSION AND FUTURE WORK}

We presented a checkpointed, quantum-resilient file transfer protocol for disrupt

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\end{thebibliography}  
\endgroup  
  
\end{document}
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